

Asymptotic analysis on autoignition and explosion limits of hydrogen–oxygen mixtures in homogeneous systems

Yongfeng Liu^a, Pucheng Pei^{b,*}

^a*Mechanical and Electrical Engineering Department, Beijing Institute of Civil Engineering and Architecture, Beijing 100044, People's Republic of China*

^b*State Key Laboratory of Automotive Safety and Energy, Tsinghua University, Beijing 100084, People's Republic of China*

Received 23 July 2004; received in revised form 22 April 2005; accepted 9 May 2005

Available online 1 July 2005

Abstract

The deduced equations about chemical species and temperature are presented to calculate hydrogen–oxygen ignition delay time. Steady-state assumptions for many of the intermediate species are introduced to derive a new simplified 3-step mechanism. The simplified 3-step mechanism for hydrogen–oxygen leads the steady-state assumptions to linear differential equations. The competition among the full mechanism containing 17-, 8- and the simplified 3-step mechanisms is carried out. The resulting closed form solutions describe the low-, the intermediate- and the high-temperature ignition regimes and obtain an “S-shaped curve”. Finally, the two parameters on ignition and extinction of the continuously stirred flow reactor are discussed in detail and the temperature error analysis is given. It reduces the computational costs and supplies theory and methods for understanding autoignition and explosion limits of hydrogen–oxygen mixtures in homogeneous systems.

© 2005 International Association for Hydrogen Energy. Published by Elsevier Ltd. All rights reserved.

Keywords: Ignition delay time; Steady-state assumptions; Reduced mechanism; Explosion limits

1. Introduction

Clean combustion has become an important aspect in the last few decades. Besides heat, combustion of hydrocarbon fuel always generates unwanted products or pollutants. Some examples of pollutant species are soot and carbon monoxide (CO), which is highly toxic, and sulphur and nitrogen oxides (SO_x and NO_x), which cause acid rain [1]. However, combustion of hydrogen–oxygen produces less pollutant without CO₂ emission. The world's climate is in a dangerous situation and we must do every thing we can now to get CO₂ emission falling before the year 2050 [2]. To meet this aim a major key lies in a deeper

understanding of the hydrogen–oxygen combustion process itself [3]. Ignition, which is one of the most important parts of combustion and directly affects combustion efficiency and emission, has been the subject of many studies that try to elucidate the ignition characteristics [4]. A particular striking feature is 2-stage ignition, which is also related to cool phenomena and the negative temperature coefficient (NTC) of ignition delay time t_i [5]. The simulations of ignition delay have been performed using the SPICE-code and a 1-D unstationary Flamelet code. The chemical mechanism for hydrogen–oxygen employed in the flamelet code was a systematically 16-mechanism developed by Mueller [6]. This mechanism is able to describe the ignition delay time at wide ranges of temperatures, pressures, equivalence ratios and different mixture fraction gradients of the flamelet. The main difficulty consisted in solving differential equations about chemical reaction rate. Fortunately, one way

* Corresponding author. Tel.: +86 10 62789134;
fax: +86 10 627857078.

E-mail address: pchpei@tsinghua.edu.cn (P. Pei).

to overcome the difficulties introduced by the non-uniqueness of the system of differential equations is the truncation of some steady-state relations [7].

The first and probably the most well-known systematic reduction method is introduced by Peters et al. [8]. This method is called steady-state solutions, which make the corresponding differential equation to be replaced by an algebraic relation. Reduced mechanisms are useful for at least two applications [9]: (1) they reduce the computational effort in numerical calculations of flames by replacing the differential equations for those intermediate species that are assumed as being in steady-state by algebraic relations; (2) they allow us to study the flame structure by asymptotic methods and thereby help to identify the relatively few kinetic parameters that mainly influence global properties such as the burning velocity or extinction strain rates. In contrast to a previous attempt by Scheid [10], where a 4-step ad hoc mechanism based on activation energy arguments was postulated and its performance was analyzed a posteriori, we will systematically derive a new 3-step reduced mechanism using steady-state approximations. This has the advantage that the competition between chain branching and chain breaking can be quantified on the basis of elementary reaction rates. Besides this, we will also consider the case of a well-stirred reactor, which has a continuous in-flow of reactants and out-flow of products. Using steady-state assumptions, we obtain ignition temperature regimes and can further calculate ignition delay time. These simplified chemistry models reduce the computational effort considerably without losing too much accuracy and predict the ignition character. The method in the paper can not only be used for hydrogen–oxygen flame, but also for any other fuels such as *n*-heptane, *n*-decane and *a*-methyl-naphthalene, etc.

2. Deduction for theory of thermal explosion

It is convenient to assume that during the ignition process either the density or the pressure remains constant. Here, we want to consider the constant density case. Let us consider a closed adiabatic vessel at constant volume V containing a given mass m such that the density $\rho = m/V$ remains constant during the ignition process as shown in Fig. 1.

With $dq=0$ and neglecting the frictional work, $dw_R=0$, the first law of thermodynamics describes that the change of internal energy vanishes:

$$\frac{du}{dt} = 0 \quad (1)$$

with $\rho = \text{const}$ the equations for the chemical species are written as [11]

$$\rho \frac{dY_i}{dt} = W_i \sum_{k=1}^r v_{ik} \omega_k. \quad (2)$$

With (2) this leads to the temperature equation

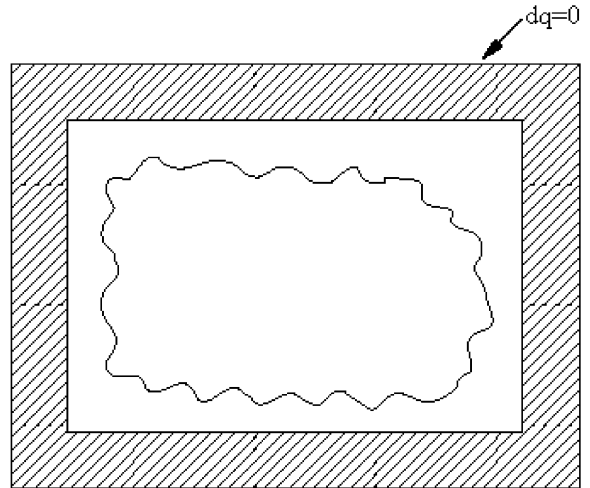


Fig. 1. Autoignition in a closed volume ($\rho = \text{const}$).

$$\rho c_v \frac{dT}{dt} = \sum Q_{vk} \omega_k, \quad (3)$$

where

$$Q_{vk} = - \sum_{i=1}^n v_{ik} W_i u_i \quad (4)$$

is the heat of combustion for a constant volume process; c_v is the mean specific heat capacity at constant volume.

Let us, for simplicity, assume a 1-step reaction with a large activation energy E and the reaction rate

$$\omega = B \left(\frac{\rho Y_F}{W_F} \right) \left(\frac{\rho Y_{O_2}}{W_{O_2}} \right) \exp \left(- \frac{E}{RT} \right). \quad (5)$$

The governing equations then simplify to

$$\rho \frac{dY_i}{dt} = v_i W_i \omega, \quad (6)$$

$$\rho c_v \frac{dT}{dt} = Q_v \omega. \quad (7)$$

Assuming constant c_v and Q_v and combining (6) and (7), one obtains

$$\frac{c_v(T - T_0)}{Q_v} = \frac{(Y_i - Y_{i0})}{v_i W_i}, \quad (8)$$

where T_0 is the initial temperature before thermal runaway and Y_{i0} is the initial mass fraction.

The temperature is expanded as

$$T = T_0(1 + \varepsilon y), \quad (9)$$

where ε is a small parameter.

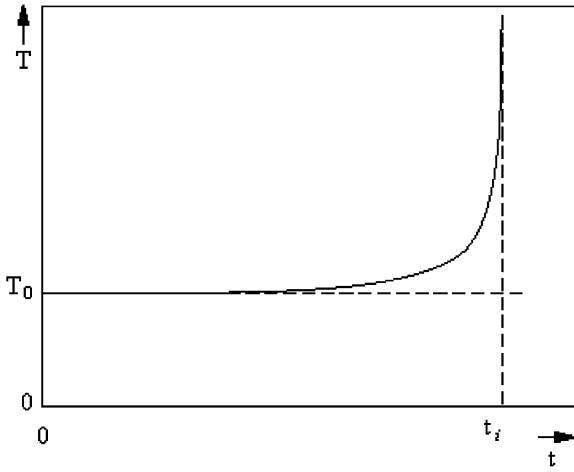


Fig. 2. Ignition delay time and temperature.

A Taylor expansion of the inverse of the temperature [12]

$$\frac{1}{T} = \frac{1}{T_0} - \frac{T - T_0}{T_0^2} = \frac{1}{T_0} (1 - \varepsilon y) \quad (10)$$

leads to an expansion of the exponential term in (5) as

$$\exp\left(-\frac{E}{RT}\right) = \exp\left(-\frac{E}{RT_0}\right) \exp\left(\frac{E}{RT_0} \varepsilon y\right). \quad (11)$$

This suggests the definition of ε as

$$\varepsilon = \frac{RT_0}{E}, \quad (12)$$

which implies that the activation energy is as large as it was assumed above. When the expansion (9) together with (11) is introduced into (7) one obtains

$$\frac{dy}{dt} = \frac{e^y}{t_i}. \quad (13)$$

The ignition delay time is defined as

$$t_i = \frac{RT_0^2}{E} \frac{c_v}{Q_v B'} \exp\left(\frac{E}{RT_0}\right), \quad (14)$$

where

$$B' = B \frac{Y_{F,0}}{W_F} \frac{\rho Y_{O_2,0}}{W_{O_2}}. \quad (15)$$

The solution of (13) is with the initial condition $y=0$ at $t=0$ using the transformation $x = e^{-y}$ which leads to $dx/dt = -1/t_i$ with the solution $x = 1 - t/t_i$. In terms of temperature T , we have

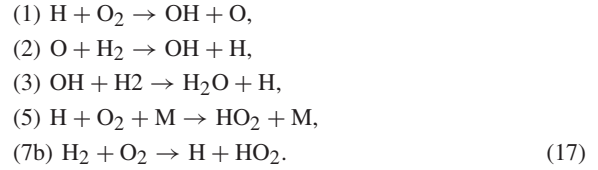
$$T = T_0 - \frac{RT_0^2}{E} \log\left(1 - \frac{t}{t_i}\right), \quad (16)$$

which is plotted in Fig. 2 and shows a thermal runaway at $t = t_i$. The solution is valid only as long as RT_0/E is small

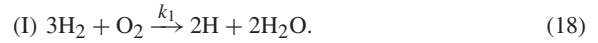
and T does not exceed the order of $T_0(1 + RT_0^2/E)$, since y was assumed to be of order unity.

3. Ignition of hydrogen–oxygen mixtures

In order to analyze the competition between chain-branching and chain-breaking reactions we will consider a simplified mechanism for the ignition of hydrogen–oxygen or hydrogen–air mixtures:



Assuming reactions (2) and (3) are very fast and thereby O and OH in steady state, we can combine the first three reactions to a global step.



As it is easily calculated the heat of combustion of reaction (I) is 48 kJ/mol while reaction (7b) is slightly endothermic with -21 kJ/mol. Reaction (5) releases most of the heat with 197 kJ/mol. Only this contribution will be retained in the temperature equation that then reads

$$\rho c_v \frac{dT}{dt} = Q_{v5} \omega_5. \quad (19)$$

Therefore a significant change of temperature will only occur when reaction (5) becomes important. The balance equation for the formation of H-radicals is

$$\rho \frac{dY_H}{dt} = W_H(2\omega_I + \omega_{7b} - \omega_5). \quad (20)$$

It will be assumed for simplicity that during the first stage of ignition, the temperature change is small and the reactions and the rate coefficients remain constant. Introducing non-dimensional variables

$$\begin{aligned} x &= \frac{Y_H k_1(T_0)}{(Y_{H_2})_0 k_{7b}(T_0)}, \\ \tau &= t k_1(T_0) \rho_0 Y_{O_2,0} / W_{O_2}, \\ \kappa &= k_5(T_0) z_5 p / (k_1(T_0) R T_0), \end{aligned} \quad (21)$$

(20) becomes

$$\frac{dx}{d\tau} = (2 - \kappa)x + 1 \quad (22)$$

with the initial conditions $x = 0$ at $\tau = 0$. The solution is

$$x = \frac{\exp[(2 - \kappa)\tau] - 1}{2 - \kappa}. \quad (23)$$

The non-dimensional radical concentration is plotted for $\kappa = 0.1, 2$ and 10 in Fig. 3. This shows that it increases

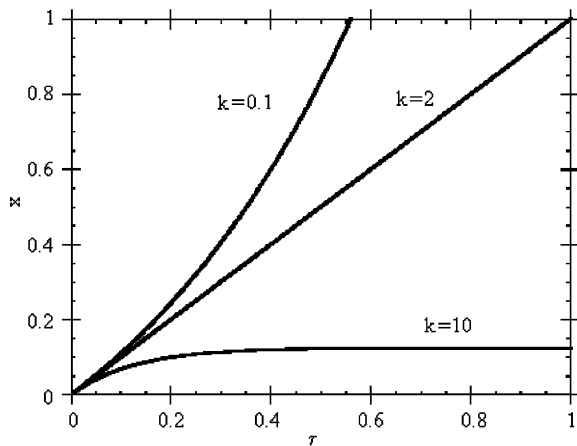


Fig. 3. Non-dimensional radical concentration.

exponentially if $k < 2$, linearly for $k = 2$ and reaches a constant value $x = 1/(k - 2)$ for $k > 2$.

While (16) readily suggests a definition of the ignition delay time as $t = t_i$, where $y \rightarrow \infty$, such a definition is not evident for solution (23). Here the initially exponential growth of the radical concentration for $k < 2$ and the existence of a threshold at $k = 2$ are the main results. The condition $k = 2$ defines a cross-over temperature T_c between the first and the fifth reactions

$$2k_1(T_c) = \frac{z_5 P}{RT_c} k_5(T_c). \quad (24)$$

Here z_5 is the effective third body efficiency

$$z_5 = X_{H_2,0} + 0.4X_{O_2,0} + 0.4X_{N_2,0} \quad (25)$$

of the initial fuel–air mixture. It is around 0.6 for stoichiometric hydrogen–air flames and around 0.8 for stoichiometric hydrogen–oxygen flames. Table 1 shows the constants in the expression $k = BT^n \exp(-E/RT)$ in the rate coefficients of the important elementary reactions of hydrogen–oxygen mixtures. The pressure dependence of T_c was evaluated using $z_5 = 0.7$ and the rates from Table 1 and is plotted in Fig. 4 below. For $p = 1$ atm T_c is close to 1000 K. Below this temperature the chain-breaking effect of reaction (5) dominates as compared to the chain-branching effect of reaction 1.

For a homogeneous stoichiometric hydrogen–oxygen mixture at 0.1 bar the onset of ignition was calculated using a 17-step mechanism in Table 1, the first 8-step mechanism and the simplified 3-step mechanism of reactions (5), (7b) and (I). Ignition delay time for both cases are calculated numerically and are compared at different initial temperature and are shown in an Arrhenius diagram in Fig. 4. It is seen that the ignition delay time increases with decreasing temperature in the range from 1500 to around 800 K. Then, at 790 K which corresponds to the cross-over temperature T_c at $p = 0.1$ bar, it suddenly increases to very large values, making ignition in technical systems quasi-impossible.

Table 1
Reactions rate coefficients constants

No.	Reaction	B	n	$E(KJ)$
1f	$H + O_2 \rightarrow O + OH$	$2.00E+14$	0.00	70.30
1b	$O + OH \rightarrow H + O_2$	$1.58E+13$	0.00	3.52
2f	$O + H_2 \rightarrow OH + H$	$5.06E+04$	2.67	26.30
2b	$OH + H \rightarrow O + H_2$	$2.22E+04$	2.67	18.29
3f	$H_2 + OH \rightarrow H_2O + H$	$1.00E+08$	1.60	13.80
3b	$H_2O + H \rightarrow H_2 + OH$	$4.31E+08$	1.60	76.46
4f	$OH + OH \rightarrow H_2O + O$	$1.50E+08$	1.14	0.42
4b	$H_2O + O \rightarrow OH + OH$	$1.47E+10$	1.14	71.09
5	$H + O_2 + M \rightarrow HO_2 + M$	$2.30E+18$	−0.8	0.00
6	$H + HO_2 \rightarrow OH + OH$	$1.50E+14$	−0.8	4.20
7	$H + HO_2 \rightarrow H_2 + O_2$	$2.50E+13$	0.00	2.90
8	$OH + HO_2 \rightarrow H_2O + O_2$	$6.00E+13$	0.00	0.00
9	$HO_2 + H \rightarrow HO_2 + O$	$3.00E+13$	0.00	7.20
10	$HO_2 + O \rightarrow OH + O_2$	$1.80E+13$	0.00	−1.70
11	$HO_2 + H_2O \rightarrow H_2O_2 + O_2$	$2.50E+11$	0.00	−5.20
12f	$OH + OH + M \rightarrow H_2O_2 + M$	$3.25E+22$	−2.0	0.00
12b	$H_2O_2 + M \rightarrow OH + OH + M$	$1.69E+24$	−2.0	202.3
13	$H_2O_2 + H \rightarrow H_2O + OH$	$1.00E+13$	0.00	15.00
14f	$H_2O_2 + H \rightarrow H_2O + H_2$	$1.70E+12$	0.00	15.70
14b	$H_2O + H_2 \rightarrow H_2O_2 + H$	$1.15E+12$	0.00	80.88
15	$H + H + M \rightarrow H_2 + M$	$1.80E+18$	−1.0	0.00
16	$OH + H + M \rightarrow H_2O + M$	$2.20E+22$	−2.0	0.00
17	$O + O + M \rightarrow O_2 + M$	$2.90E+17$	−1.0	0.00

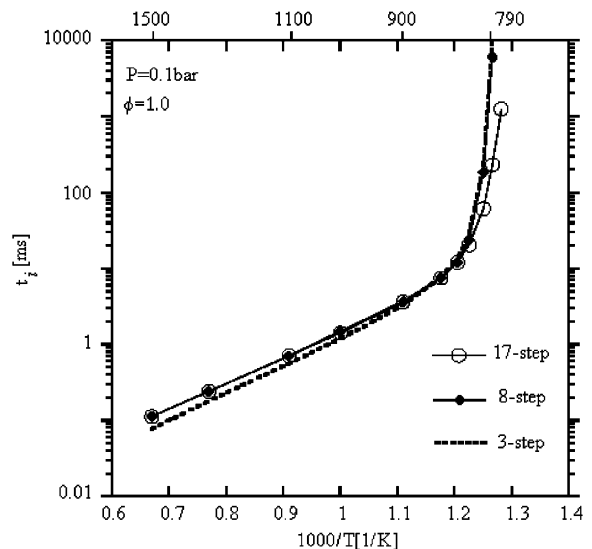


Fig. 4. Ignition delay time for stoichiometric hydrogen–oxygen mixtures.

4. Explosion limits of hydrogen–oxygen mixtures

There are conditions of pressure and temperature where the ignition delay time tends to infinity and autoignition is chemically inhibited. These conditions are called explo-

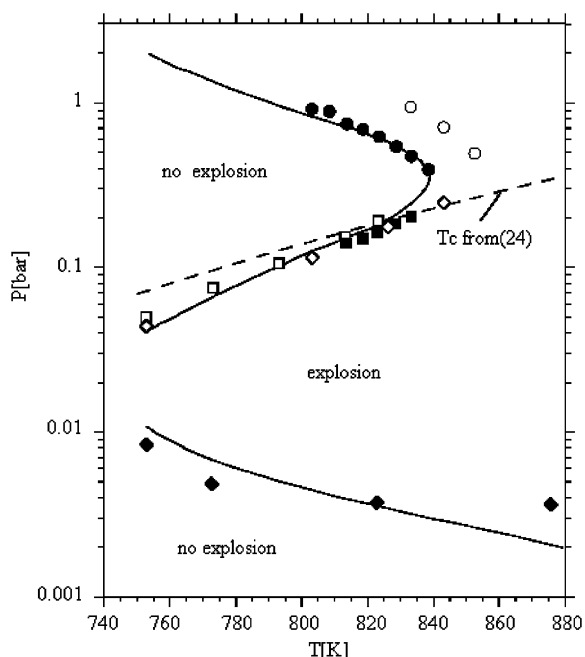


Fig. 5. Explosion limits for a stoichiometric hydrogen–oxygen mixture in a spherical vessel.

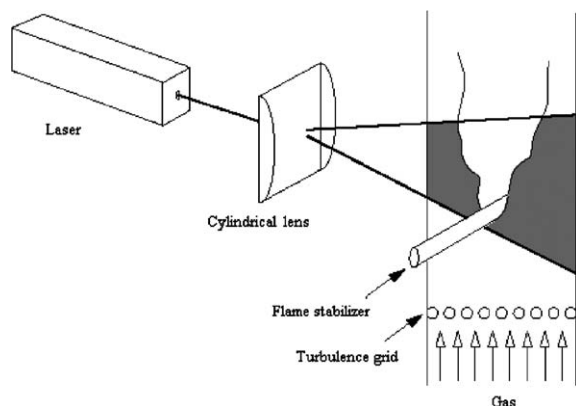


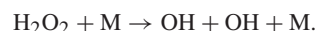
Fig. 6. Schematic of the experimental set-up.

sion limits [13]. The onset of a homogeneous explosion in a closed vessel depends on additional influences; in particular, the heat loss to the walls and thereby on the dimensions of the vessel. Fig. 5 shows the explosion limits for stoichiometric hydrogen–oxygen mixture as a function of pressure and temperature. (Solid line: numerical calculation using a full kinetic mechanism in Table 1; dashed line: (24) with $z_5 = 0.7$; point: experimental data.) Fig. 6 shows a schematic of the test bench configuration. The instantaneous flame front can be visualized by using the laser sheet technique. The beam of the laser is widened by a cylindri-

cal lens. The gas from the burner passes through a turbulent grid. The V-shaped flame is stabilized by a rod normal to the flow. Therefore, the instantaneous flame contour is then visualized as the interface between the bright shining region of the unburnt gas and the dark region of burnt gas. It may be recorded either by high-speed cinematography or by a CCD camera and then be processed and digitalized.

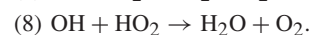
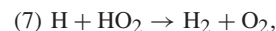
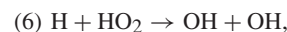
For a time range between 750 and 840 K there are three branches separating non-explosive to explosive conditions. The low- and the high-pressure branches are influenced by the depletion of chain carrier on the walls of the vessel. A different coating with KCl gives different values for the explosion limits. The middle branch is essentially determined by the competition between reaction (1) and (5). It agrees well with T_c evaluated from (24).

The lower branch may be explained as follows: For low temperatures, the initial chain branching proceeds through the reaction:

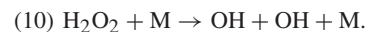
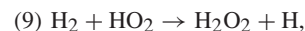


Since the first of this reaction is trimolecular with large activation energy, it increases relative to the other bimolecular reactions as the pressure increases but decreases with temperature. A line of constant rate should then have a negative slope in the p – T Fig. 5. The explosion limits are then given by the condition that the rate of removal of radicals by wall reactions equals that of production by the two reactions above.

For pressures above the middle branch in Fig. 5, chain braking by reaction (5), leads to a massive formation of HO_2 . This reacts with H and OH as



Only the first of these is radical conserving, while reactions (7) and (8) are chain breaking. Therefore, HO_2 is not a chain carrier at typical combustion conditions. When a large amount of HO_2 is being formed, however, there is a chain branching effect through the reactions



Whether chain branching by reactions (9) and (10) dominates over chain breaking by reactions (7) and (8) depends on the ratio of H and OH versus H_2 and on temperature. In addition, the possibility of chain branching in a closed vessel depends again on the vessel diameter. The larger the vessel and therefore the volume to wall area ratio, less radicals are removed compared to those that are being formed. Therefore, an increase of pressure beyond the second explosion limit will eventually lead to the third explosion limit, the upper branch in Fig. 5. Since reactions (9) and (10) increase

with temperature, the third branch is reached later at lower temperatures, which explains its negative slope in Fig. 5.

5. Ignition and extinction of the continuously stirred flow reactor

We now consider an idealized flow reactor of constant mass m and volume V as shown in Fig. 7. The mass flow rate $\dot{m}$ enters with reactants $Y_i = Y_{i,u}$ at temperature $T = T_u$. The reactants are immediately mixed with combustion products at the entrance of the reactor where the flow is assumed to be highly stirred by many incoming turbulent jets.

The temperature and the concentrations within the reaction are assumed to be uniform equal to T and Y_i , respectively. The reactor is assumed adiabatic for simplicity.

The species equation [14]:

$$m \frac{dY_i}{dt} = \dot{m}(Y_{i,u} - Y_i) + V W_i \sum_{k=1}^r v_{ik} \omega_k, \quad (26)$$

the temperature equation:

$$m c_p \frac{dT}{dt} = \dot{m} c_p (T_u - T) + V \sum_{k=1}^r Q_k \omega_k. \quad (27)$$

In (15) B has the dimension s^{-1} and therefore represents the inverse of a reaction time. It must be compared to the residence time t_v within the reactor which is

$$t_v = \frac{m}{\dot{m}}. \quad (28)$$

Using $\rho = m/V$ and introducing the non-dimensional quantities

$$Da = B t_v, \quad Y^* = \frac{Y_F}{Y_{F,u}}, \quad T^* = \frac{T}{T_u}, \quad (29)$$

$$t^* = \frac{t}{t_v}, \quad E^* = \frac{E}{R T_u}, \quad Q^* = \frac{Q Y_u}{c_p W_F T_u},$$

One obtains the non-dimensional reactor equations (with the asterisk removed)

$$\frac{dY}{dt} = 1 - Y - Da Y \exp(-E/T),$$

$$\frac{dT}{dt} = 1 - T - Q Da Y \exp(-E/T). \quad (30)$$

The parameters of these equations are the Damkohlers number Da , the non-dimensional heat of combustion Q and the non-dimensional activation energy E .

$$\text{Introducing } T + QY = 1 + Q \quad (31)$$

into (30) the problem of the adiabatic reactor is described by a single equation for the temperature

$$\frac{dT}{dt} = 1 - T + Da N(T), \quad (32)$$

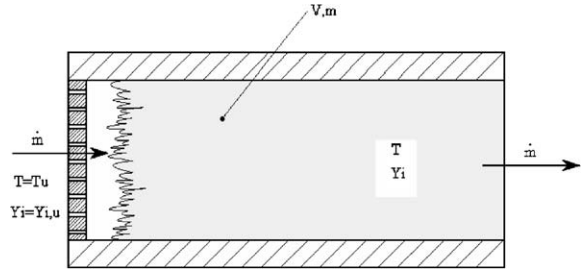


Fig. 7. Schematic representation of a continuously stirred flow reactor.

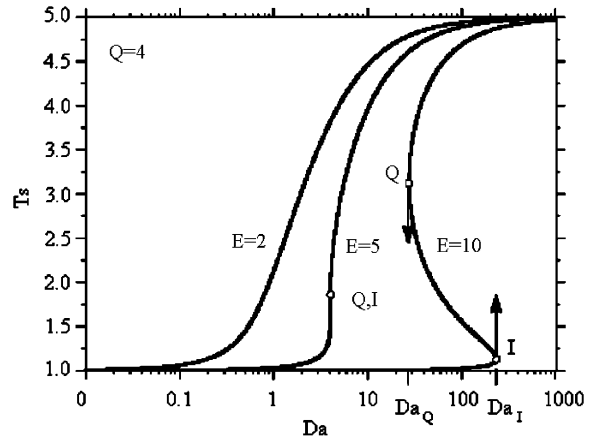


Fig. 8. Steady-state solution of the reactor equations.

where the non-linear operator N is defined by

$$N = (1 + Q - T) \exp(-E/T). \quad (33)$$

We will now assume that the reactor has reached a steady-state. The steady-state temperature T_s is obtained from the algebraic equation

$$T_s + Da N(T_s) = 1. \quad (34)$$

Solutions of (34) for $Q=4$ and $E=2, 5$ and 10 are shown in Fig. 8. For $E=2$ one obtains a monotonous increase of T_s with increasing Damkohler number. For $E=10$, however, there is a range of Damkohler numbers between Da_Q and Da_I where multiple solutions exist. The shape of this curve reminds us vaguely of an S and is often called the “S-shaped curve”. At the points Q and I , corresponding to the Damkohler number Da_Q and Da_I the curve for $E=10$ has a vertical tangent. If the reactor is originally operated with a low residence time and therefore a small Damkohler number, the solution remains close to $T_s = 1$ corresponding to the non-reacting case. When the Damkohler number increases beyond $Da = 100$ in the case $E=10$, there is a small temperature increase. Beyond $Da = Da_I$, however, the low-temperature solution no longer exists and the only

steady-state solution at which the reactor can operate is the high-temperature branch. The unstationary transition from the low- to the high-temperature branch corresponds to an ignition process. It starts from the point I of the lower temperature branch, and is denoted by an arrow in Fig. 8.

When the upper branch is reached, steady-state combustion is established. If one then reduces the residence time again, thereby lowering the Damkoehler number, one follows the curve $E=10$ to the left towards lower temperatures. This indicates that the available residence time is insufficient for complete combustion and that an increasing amount of fuel will remain unreacted. When the Damkoehler number becomes lower than Da_Q , an unstationary transition from the upper to the lower temperature branch occurs. This corresponds to the extinction or quenching of the reaction. The temperature then drops from the value at the point Q to a value close to $T=1$. This is again denoted by an arrow in Fig. 8. The ignition at a large Damkoehler number Da_I and subsequent extinction at a lower Damkoehler number Da_Q is a typical hysteresis process. We will show below that only the upper and the lower branches of the steady-state solution are stable while the middle branch solution is unstable.

6. Discussion

6.1. Two values of T

In order to determine the conditions for which multiple solution exist, we want to calculate the temperature T_I and T_Q corresponding to the ignition and quenching points. At these points the curve has a vertical tangent leading to the condition

$$\frac{dDa}{dT_S} = 0. \quad (35)$$

Since (34) may be expressed as

$$Da = \frac{T_S - 1}{N(T_S)}, \quad (36)$$

(35) leads to

$$N(T_S) - (T_S - 1) \frac{dN(T_S)}{dT_S} = 0 \quad (37)$$

with

$$\frac{dN(T_S)}{dT_S} = N(T_S) \left(\frac{E}{T_S^2} - \frac{1}{1 + Q - T_S} \right), \quad (38)$$

one obtains after some algebraic manipulation a quadratic equation which has the solution

$$T_{I,Q} = \frac{(2 + Q) \pm Q\sqrt{1 - 4(1 + Q)/EQ}}{2(1 + Q/E)}. \quad (39)$$

The term under the square root vanishes for $E=4(1+Q)/Q$. For the value $Q=4$ chosen in Fig. 8 this corresponds to $E=5$.

Two solutions and therefore ignition as well as quenching exist only if

$$E > 4(1 + Q)/Q. \quad (40)$$

For smaller values of the activation energy a monotonous transition from lower to higher temperatures takes place. This is shown for $E=2$ in Fig. 8. Typical combustion processes starting from ambient temperature are nearly always in the regime of large non-dimensional activation energies for the temperatures T_I and T_Q resulting from (39) may be obtained in the limit $E \rightarrow \infty$. Expanding the square root term and the term in the denominator of (39) up to first order as

$$\begin{aligned} \sqrt{1 - 4(1 + Q)/EQ} &\approx 1 - 2(1 + Q)/EQ, \\ (1 + Q/E)^{-1} &\approx 1 - Q/E, \end{aligned} \quad (41)$$

one obtains

$$T_I = 1 + 1/E, \quad T_Q = 1 + Q - \frac{(1 + Q)^2}{E}. \quad (42)$$

In terms of dimensional quantities this may be expressed as

$$\begin{aligned} T_I &= T_{\min} \left(1 + \frac{RT_{\min}}{E} \right), \\ T_Q &= T_{\max} \left(1 - \frac{RT_{\max}}{E} \right). \end{aligned} \quad (43)$$

It is easily seen that for ignition the temperature increment $RT_{\min}/E$ corresponds to that used in (9).

6.2. Two cases of ε

It is worthwhile to study the upper branch of the steady-state solution up to the extinction point by an asymptotic expansion. We write the non-dimensional temperature as

$$T_S - Q + \varepsilon y = 1, \quad (44)$$

where ε is yet to be determined. Expanding the inverse temperature in the exponential term in (33) as

$$\frac{1}{T_S} = \frac{1}{(1 + Q)[1 - \varepsilon y/(1 + Q)]} = \frac{1}{1 + Q} + \frac{\varepsilon y}{(1 + Q)^2} \quad (45)$$

and inserting this into (34) one obtains

$$Q - \varepsilon y - Da \exp\left(-\frac{E}{1 + Q}\right) \exp\left(-\frac{E\varepsilon}{(1 + Q)^2} y\right) = 0 \quad (46)$$

in the limit $\varepsilon \rightarrow 0$ the second term in this equation vanishes. Assuming Q of order unity, a balance of the first with the third term then requires that the Damkoehler number is sufficiently large. We shall now discuss (46). Depending on the magnitude of the activation energy, we have the two following cases.

6.2.1. The non-dimensional activation energy E is of order unity

This suggests the definition $\varepsilon = 1/Da$ leading to a large Damkoehler number analysis. The second exponential then vanishes in the limit $\varepsilon \rightarrow 0$ and with

$$y = Q \exp\left(\frac{E}{1+Q}\right), \quad (47)$$

the temperature expansion (44) becomes

$$T_S = 1 + Q - \frac{Q}{Da} \exp\left(\frac{E}{1+Q}\right). \quad (48)$$

This expression is able to approximate the upper branch of the S-shaped curve for the case $E = 2$ in Fig. 8. It is not suitable for the case $E = 10$ and is in particular not able to predict the turning point of this curve at extinction.

6.2.2. The non-dimensional activation energy E is asymptotically large

The grouping in the second exponential in (46), then suggests the definition

$$\varepsilon = \frac{(1+Q)^2}{E} \quad (49)$$

equivalent to large activation energy asymptotes. In order to obtain a balance of the remaining terms in (46) one must require that the quantity

$$\delta = \frac{Da(1+Q)^2}{EQ} \exp\left(-\frac{E}{1+Q}\right) \quad (50)$$

is of order unity. This indicates that the Damkoehler number must grow as $Da \propto \varepsilon^{-1} \exp((1+Q)/\varepsilon)$ in the limit $\varepsilon \rightarrow 0$. This linking of two asymptotically large parameters in a specific way is called a distinguished limit. Inserting (49) and (50) into (46) leads to the nonlinear equation of

$$1 = \delta y \exp(-y) \quad (51)$$

from which y may be determined. Its solution is plotted in Fig. 9, showing a turning point at $y = 1$, $\delta = e$. For values of $\delta > e$ there are two solutions, the lower one corresponding to the upper branch in Fig. 9, the upper one corresponding to the intermediate branch. The turning point characterized the extinction condition. It shall be noted that the analysis of extinction requires large activation energy rather than large Damkoehler number asymptotic.

6.3. Error analysis

Finally, we want to show that, when multiple solutions exist, only the lower and the upper branches of the S-shaped curve are stable, while the intermediate branch is unstable. We go back to (32) and introduce the small temperature perturbation T' [15] of the steady-state solution

$$T = T_S + T'(t). \quad (52)$$

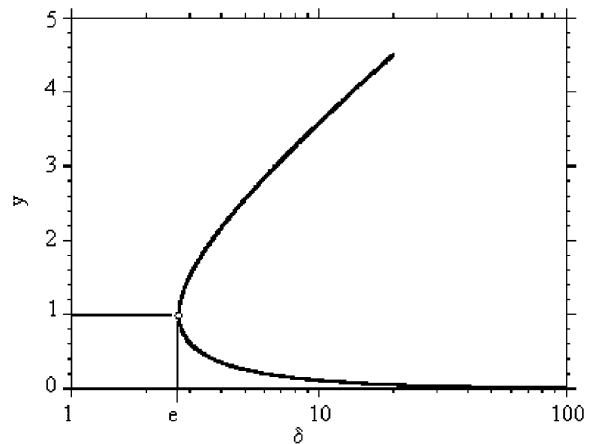


Fig. 9. Large activation energy asymptotic solution.

When this is introduced into (32) and (34) is subtracted, one obtains after linearization

$$\frac{dT'}{dt} = -\lambda T', \quad (53)$$

where

$$\lambda = 1 - Da \frac{dN(T_S)}{dT_S} = N(T_S) \frac{dDa}{dT_S}. \quad (54)$$

Here (34) has been used. The perturbation

$$T' = T'_0 \exp(-\lambda t), \quad (55)$$

where T'_0 is an arbitrary initial perturbation, clearly decays with time if it is positive, but it grows if λ is negative. Since the sign of λ equals the sign of dDa/dT_S , steady-state solutions showing an increase of temperature with increasing Damkoehler number are stable. This is true for the lower and the upper branches of the S-shaped curve. The opposite is true for the middle one which is therefore unstable.

7. Conclusions

By analyzing mathematically and using steady-state assumptions, we were able to identify the rate determining reaction in the three regimes of ignition in closed volume.

(1) For a homogeneous stoichiometric hydrogen–oxygen mixture at 0.1 bar, the ignition delay time increases with decreasing temperature in the range from 1500 to around 800 K.

(2) In a closed volume, there are three branches separating non-explosive to explosive conditions for a time range between 750 and 840 K. The lower and the upper branches are stable and the middle branch is unstable.

(3) In the continuously stirred flow reactor, the ignition regime of the intermediate temperature is unstable, and

the regimes of the high and the low temperatures are stable. The activation energy decides the conditions of ignition and extinction in homogeneous systems rather than the Damkoehler number.

Acknowledgements

Prof. Dr.-Ing. Dr. H.C. Dr.-Ing. E.H. Norbert Peters, who is a member of Germany Science Academy and a Foreign Associate of the US National Academy of Engineering, gave some good advices about the paper and the Ford Research Center in Aachen, Germany, supplied experiment equipments and colleagues with experiments about the paper. This project has been supported by China Scholarship Council (No. 21811213).

References

- [1] van Oijen JA. Modelling of premixed burner systems by using flamelet-generated manifolds. *Combust Flame* 2001;127: 2124–34.
- [2] World Energy Assessment. Energy and the challenge of sustainability. United Nations development programme, New York, 2000.
- [3] Peters N. Turbulent combustion. Cambridge: Cambridge University Press; 2004.
- [4] Bongers H. Intrinsic low-dimensional manifold method extended with diffusion. *Proc Combust Inst* 2002;29.
- [5] Hasse C, Bikas G, Peters N. Modeling DI-diesel combustion using the Eulerian particle flamelet model. SAE 2000-01-2934, 2000.
- [6] Mueller UC, Peters N. Asymptotic analysis for premixed hydrogen flame. In: Twenty-fourth international symposium on combustion. Pittsburgh, 1992. p. 777–84.
- [7] Hergart C, Barths H, Peters N. Modeling the combustion process in a small-bore diesel engine using a model based on representative interactive flamelets. SAE 1999-01-3550, 1999.
- [8] van Oijen JA. Modeling of premixed counterflow flames using the flamelet-generated manifold method. *Combust Theory Model* 2002;6:463–78.
- [9] Peters N, Rogg B, editors. Reduced kinetic mechanisms for combustion applications. Berlin: Springer; 1993.
- [10] Scheid F, Trilken B. Soot formation in combustion-mechanisms and models. Berlin-Heidelberg: Springer; 1994.
- [11] Pope SB. Computationally efficient implementation of combustion chemistry using in adaptive tabulation. *Combust Theory Model* 1997;1:41–63.
- [12] Groot GRA, de Goey LPH. A computational study on propagating spherical and cylindrical premixed flames. *Proceed Combust Inst* 2002;29..
- [13] Carl-Anders Hergart. Modeling combustion and soot emissions in a small-bore direct-injection diesel engine. Aachen: Shaker Verlag; 2001. p. 10–22.
- [14] Barths H. Simulation of diesel engine and gas turbine combustion using multiple flamelets with detailed chemistry. Aachen: Shaker Verlag; 2000. p. 31–41.
- [15] Williams FA. In: Buckmaster J, editor. Turbulent combustion in the mathematics of combustion, 1985.